

Formula

① Formation of PDE: Eliminate

arb. constant

Find derivative of z equal to number of constant

arb. function

Find derivative of z upto to order equal to number of function

② Lagrange method

$$Pp + Qq = R$$

Auxiliary equation

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

③ Charpit method

Non Linear 1st order PDE

Ex: $px + qy = pq, p^2 + q^2 = 1$

A. eqn

$$\frac{dp}{p + p_1 p_2} = \frac{dq}{q + q_1 q_2} = \frac{dz}{-p_1 p_2 - q_1 q_2}$$

$$= \frac{dx}{-p_1} = \frac{dy}{-q_1} = \frac{dF}{0}, \text{ Find its}$$

one solution containing p or q . Find value of p & q using this solution & given PDE. Put p, q in $dz = p dx + q dy$

④ Linear Homogeneous PDE of higher order with constant coefficient:

$$F(D, D')z = f(x, y)$$

A CF

① $A \cdot e^{mx}$ $F(m, 1) = 0$

② Find roots

③ If distinct say a, b

$$CF = f_1(y + ax) + f_2(y + bx)$$

④ If repeated a, a

$$CF = f_1(y + ax) + x f_2(y + ax)$$

Sheet Unit-1

B $PI = \frac{1}{F(D, D')} f(x, y)$

Case-1 $f(x, y) = \phi(ax + by)$

$$PI = \frac{1}{F(a, b)} \iint \phi(u) du du \dots$$

Where (n) is order of operator such that $F(a, b) \neq 0$

Note: If $F(a, b) = 0$

$$PI = x \left[\frac{1}{F'(D, D')} f(x, y) \right]$$

Derivative w.r.t D

Case-2 $f(x, y) = x^n y^m$

$$PI = \frac{1}{F(D, D')} x^n y^m$$

① $F(D, D') \rightarrow 1 \pm \phi\left(\frac{D}{D'}\right)$ OR $\frac{D'}{D}$

② $\frac{1}{F(D, D')} \rightarrow \left[1 \pm \phi\left(\frac{D}{D'}\right)\right]^{-1}$

③ Expand upto degree n or m taking highest degree term of D in common

Case-3 General Case

① $F(D, D') \rightarrow$ Linear factor like $D \pm mD'$

② $\frac{1}{D \pm mD'} f(x, y) = \int f(x, c + mx) dx$

after $\int$ put back $c + mx = y$

If D' is common factor in $F(D, D')$ then

$$CF \text{ for } D' = f_1(x)$$

$$D'^2 = f_1(x) + y f_2(x)$$

5 Linear Non homogeneous PDE of higher order with constant coefficient

$$F(D, D')Z = f(x, y)$$

(A) CF (1) $F(D, D') \rightarrow$ Linear factor

(2) $D - mD' - a \rightarrow e^{ax} f(y + mx)$

$(D - mD' - a)^2 \rightarrow e^{ax} f_1(y + mx) + x e^{ax} f_2(y + mx)$

(3) $D' - a \rightarrow e^{ay} f(x)$

(4) $(D' - a)^2 \rightarrow e^{ay} f_1(x) + y e^{ay} f_2(x)$

(B) $PI = \frac{1}{F(D, D')} f(x, y)$

Case-1 $f(x, y) = e^{ax+by}$

$$PI = \frac{1}{F(a, b)} e^{ax+by}$$

Case-2 $f(x, y) = \sin(ax+by)$ or $\cos(ax+by)$

Put $D^2 = -a^2, D'^2 = -b^2, DD' = -ab$

In $F(D, D')$

Case-3 $f(x, y) = x^n y^m$

$$PI = \frac{1}{F(D, D')} x^n y^m$$

(1) $F(D, D') \rightarrow 1 \pm \phi(D, D')$ by taking least degree term in common

$$(2) \frac{1}{F(D, D')} = [1 \pm \phi(D, D')]^{-1}$$

$\rightarrow$ expand upto degree $m+n$

No x, y, z

are connected by $f(a, b) = 0$

Case-4 $f(x, y) = e^{ax+by} \phi(x, y)$

$$PI = \frac{1}{F(D, D')} e^{ax+by} \phi(x, y) = e^{ax+by} \left[\frac{1}{F(D+a, D'+b)} \phi(x, y) \right]$$

apply any know Rule

(6) PDE reducible to homo or Non homo geneous PDE with constant coefficient

(1) EX $x^2 \frac{\partial^2 z}{\partial x^2} + 4xy \frac{\partial^2 z}{\partial x \partial y} + 5y^2 \frac{\partial^2 z}{\partial y^2} = f(x, y)$

(2) put $x = e^u, y = e^v$ then

$$x \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}, y \frac{\partial z}{\partial y} = \frac{\partial z}{\partial v} = D'z = Dz$$

$$x^2 \frac{\partial^2 z}{\partial x^2} = D(D-1)z, y^2 \frac{\partial^2 z}{\partial y^2} = D'(D'-1)z$$

$$xy \frac{\partial^2 z}{\partial x \partial y} = DD'z$$

(3) then (1) will reduce to $F(D, D')z = \phi(u, v)$

where $D \equiv \frac{\partial}{\partial u}, D' \equiv \frac{\partial}{\partial v}$

(4) solve finding CF & PI

(5) at last put back

$$e^u = x \quad \& \quad e^v = y \Rightarrow u = \log x \Rightarrow v = \log y$$

7 Some standard form

Case-1 if eqn is in form

$f(p, q) = 0$ then solution is $z = ax + by + c$ where $a \neq b$

Case-2: if eqn is in form $z = px + qy + f(p, q)$

then solution is: $z = ax + by + f(a, b)$

Note $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$, $r = \frac{\partial^2 z}{\partial x^2}$, $s = \frac{\partial^2 z}{\partial x \partial y}$, $t = \frac{\partial^2 z}{\partial y^2}$

Sheet Notes - Unit 2

- ① PDE is Hyperbolic, Parabolic, elliptic if
- $B^2 - 4AC > 0$ $= 0$ < 0

where PDE: $A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + f(x, y, u, u_x, u_y) = 0$

is the standard form of 2nd order PDE

② Method of separation of variable

Ex $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial y} + u$

Step 1 let $u = X(x) Y(y)$ Step 2 put $u = xy$ in PDE

Step 3 Find partial derivative $\frac{\partial^2(xy)}{\partial x^2} = \frac{\partial(xy)}{\partial y} + xy$

$\Rightarrow x \frac{\partial^2 x}{\partial x^2} = x \frac{\partial y}{\partial y} + xy$ Step 4 make separation of variable. $\Rightarrow$ Divide whole eqⁿ by xy .

Put x, x at one side & y, y at another side of $= \Rightarrow \frac{1}{x} \frac{\partial^2 x}{\partial x^2} = \frac{1}{y} \frac{\partial y}{\partial y} + 1$ Step 5 Above

equality will hold only for some constant k so put it equal to k . Step 6 Replace $\partial \rightarrow d$ then we get two ordinary d. eqⁿ which we can solve easily.

$$\frac{1}{x} \frac{d^2 x}{dx^2} = \frac{1}{y} \frac{dy}{dy} + 1 = k$$

ODE of second order

$$x = CF + PI$$

ODE of first order. solve by separation of variable

Step 7. put x, y in $u = xy$ Ans

③ Wave equation: (i) Displacement of string (y)

w.r.t x & t : $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$

(ii) Its solution by method of separation of variable for $k = -\beta^2$

$$y = (A \cos \beta x + B \sin \beta x) (C \cos c \beta t + D \sin c \beta t)$$

(iii) Conditions

✓ Boundary: (i) $x=0, y=0$ (ii) $x=l, y=0$

✓ Initial: (iii) $t=0, \frac{\partial y}{\partial t} = 0$ (iv) $t=0, y = f(x)$

Apply one by one to get A, B, C, D

* Don't change the sequence of conditions

④ Heat equation (i) temperature distribution

in rod is given by $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ in one dimension

(ii) Its solution by method of separation of variable for $k = 0$

$u = Ax + B \rightarrow$ Steady state temp dist OR
and for $k = -\beta^2$ x dependent only OR
independent of time

$$u = (A \cos \beta x + B \sin \beta x) e^{-c^2 \beta^2 t}$$

$\rightarrow$ transient state temp. dist. OR
time dependent temp dist.

(iii) Conditions: Type-1 Both ends are at $0^\circ C$

i) $x=0, u=0$ ii) $x=l, u=0$

(iii) $t=0, u = f(x)$

Apply one by one on solution to get A, B, β

Type-2 OR Non zero temperature, steady state + transient state EX

① find:
Initial temp which arise due to temp at ends so its eqn will be $u_0 = an + b$

Find a, b using temp at ends
(i) $x=0, u=20$
(ii) $x=l, u=30$

We get $u_0 = \frac{10x}{l} + 20$ at $t=0$

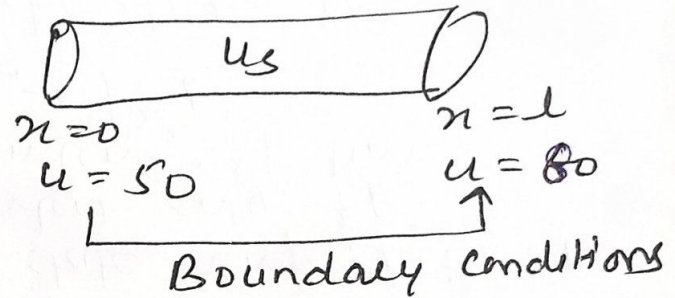
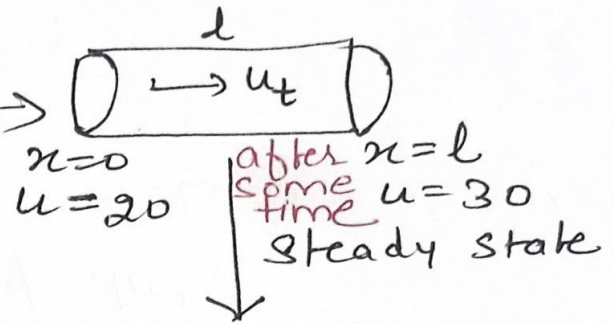
this is initial temp

② Now find steady state temp $u_s = an + b$ from the Boundary condition
(i) $x=0, u=50$ (ii) $x=l, u=80$

We get $u_s = \frac{30x}{l} + 50$

③ Also we know transient state temp
 $u_t = (A \cos pn + B \sin pn) e^{-c^2 p^2 t}$

④ So total temp dist- in the rod
 $u = u_t + u_s$



⑤ Conditions Now we have

Boundary conditions: (i) $u=50$ at $x=0$

(ii) $u=60$ at $x=l$

Initial condition (iii) at $t=0$ $u = \frac{10x}{l} + 20$

⑥ Now Apply condition one by one on (4) to find A, B, P

⑤ Laplace equation: (i) Temperature distribution in a rectangular metal plate at any point (x, y) in steady state: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ with insulated surface

(ii) Its solution by method of separation of variable for $K = p^2$
 $u = (A e^{px} + B e^{-px}) (C \cos py + D \sin py)$ I
 & for $K = -p^2$

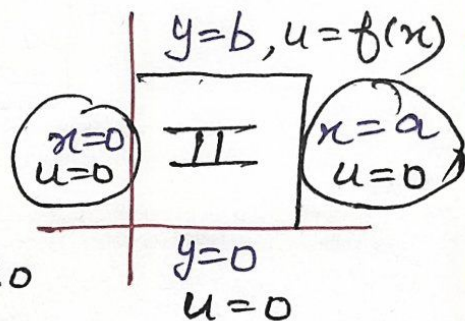
$u = (A \cos px + B \sin px) (C e^{py} + D e^{-py})$ II

Note (iii) if we have metal plate of length b (y) conditions and width a (x) then we can have:

Type-1 (i) $x=0, u=0$ (ii) $x=a, u=0$

(iii) $y=0, u=0$ (iv) $y=b, u=f(x)$

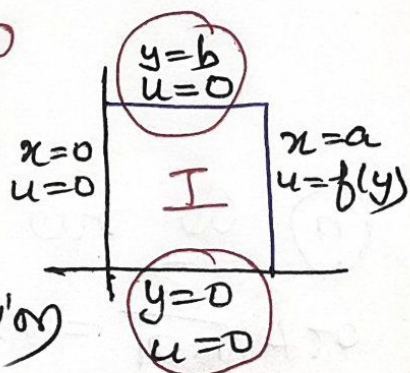
As we are given 0°C at both of the edges $x=0$ & $x=a$ so in this case take II solution



Type 2 (i) $y=0, u=0$ (ii) $y=b, u=0$

(iii) $x=0, u=0$ (iv) $x=a, u=f(y)$

In this case when edges $y=0$ & $y=b$ both are at 0°C we take I solution



Now Apply condition on solution one by one to find A, B, C, D